

A vertical shift

Use $f(x) = x^2$.

$$g(x) = f(x) + 3$$

How does the graph of g relate to the graph of f ?

A horizontal shift

Use $f(x) = x^2$.

$$g(x) = f(x - 4)$$

Where does the vertex move?

The direction of an inside shift

The point $(2, 5)$ is on the graph of f .

$$g(x) = f(x + 3)$$

Which corresponding point lies on the graph of g ?

Shift both coordinates

Use $f(x) = \sqrt{x}$.

$$g(x) = f(x - 2) - 1$$

Where does the graph begin?

Shift a domain

The domain of f is $[-2, 5]$.

$$g(x) = f(x - 3)$$

What is the domain of g ?

Reflect across the x-axis

The point (3, -2) is on the graph of f .

$$g(x) = -f(x)$$

Which corresponding point lies on the graph of g ?

Reflect across the y-axis

The point $(3, -2)$ is on the graph of f .

$$g(x) = f(-x)$$

Which corresponding point lies on the graph of g ?

Reflect a square root

Use real inputs.

$$g(x) = \sqrt{-x}$$

What does its graph look like?

An even function

A function is even when its domain is symmetric about zero and opposite inputs give equal outputs.

$$f(-x) = f(x)$$

What graph symmetry expresses this property?

An odd function

A function is odd when its domain is symmetric about zero and opposite inputs give opposite outputs.

$$f(-x) = -f(x)$$

What graph symmetry expresses this property?

Test for even symmetry

$$f(x) = x^4 + 2x^2 - 3$$

Is f even, odd, or neither?

Test for odd symmetry

$$f(x) = x^3 - 4x$$

Is f even, odd, or neither?

A function with neither symmetry

$$f(x) = x^2 + x$$

Is f even, odd, or neither?

A vertical stretch

The point $(2, 3)$ lies on the graph of f .

$$g(x) = 4f(x)$$

Which corresponding point lies on the graph of g ?

A vertical compression

Use $f(x) = |x|$.

$$g(x) = \frac{1}{2}f(x)$$

How does the graph change?

A horizontal compression

The point $(6, 2)$ lies on the graph of f .

$$g(x) = f(3x)$$

Which corresponding point lies on the graph of g ?

A horizontal stretch

The point (2, 5) lies on the graph of f.

$$g(x) = f\left(\frac{x}{4}\right)$$

Which corresponding point lies on the graph of g?

Compare inside and outside

Use $f(x) = \sqrt{x}$.

$$2f(x), \quad f(2x)$$

Do these formulas give the same function?

Combine transformations

Use $f(x) = x^2$.

$$g(x) = -2f(x - 3) + 1$$

What graph results from these transformations?

Transform a point

The point $(4, -1)$ lies on the graph of f .

$$g(x) = 3f(2(x - 1)) + 5$$

Which corresponding point lies on the graph of g ?

Factor the input first

Use $f(x) = |x|$.

$$g(x) = f(2x - 6)$$

What horizontal transformations does this formula represent?

Recover a formula

Start with $y = \sqrt{x}$. Shift left 2, reflect across the x-axis, and shift up 4.

What equation represents the transformed graph?

A transformed range

The range of f is $[-1, 4]$.

$$g(x) = -2f(x) + 3$$

What is the range of g ?

Compare two shifts

$$(x - 2)^2, \quad x^2 - 2$$

How do the two vertex locations differ?